RAJALAKSHMI ENGINEERING COLLEGE

M Hirdik 251801101-2025-AIDS

M2

Started on	Sunday, 20 September 2026, 9:55 AM
State	Finished
Completed on	Sunday, 20 September 2026, 9:57 AM
Time taken	1 min 58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void reverse(int n)
{
    int rev = 0, remainder;
    int count = 0;

    count++;  // rev = 0

    while (n != 0)
    {
        count++;  // while condition
        count++;  // remainder = n % 10
        count++;  // rev = rev * 10 + remainder
        count++;  // n /= 10

        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }

    count++;  // final while condition
    count++;  // print(rev)

    printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);

    reverse(n);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

M Hirdik 251801101-2025-AIDS

M2

Started on	Thursday, 6 August 2026, 11:47 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:55 AM
Time taken	8 mins 48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int count=0;
    int c=0;
    int n;
    scanf("%d",&n);
    count++;
    for(int i=n/2;i<n;i++){
        count++;
        for(int j=1;j<n;j=2*j){
            count++;
            for(int k=1;k<n;k=k*2){
                count++;
                count++;
                c++;
            }
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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M Hirdik 251801101-2025-AIDS

M2

Started on	Thursday, 6 August 2026, 11:17 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:16 PM
Time taken	20 days 11 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int num;
    int count=0;
    scanf("%d",&num);
    for(int i=1;i<=num;++i){
        count++;
        if(num%i==0){
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

M Hirdik 251801101-2025-AIDS

M2

Started on	Thursday, 6 August 2026, 10:34 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:05 AM
Time taken	30 mins 29 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    int count=0;
    int i,j;
    scanf("%d",&n);
    count++;
    if(n==1){
        count++;
    }
    else{
        for(i=1;i<=n;i++)
        {
            count++;
            for(j=1;j<=n;j++){
                count++;
                count++;

                count++;

                break;
            }
            count++;
        }
        count++;
    }
    printf("%d",count);
    return 0;
```

```
29 }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

M Hirdik 251801101-2025-AIDS

M2

Started on	Thursday, 6 August 2026, 10:23 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:33 AM
Time taken	10 mins 8 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

```
Convert the following algorithm into a program and find its time complexity using the counter method.
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    int i=1;
    int s=1;
    int count=2;
    scanf("%d",&n);
    while(s<=n)
    {
        count++;
        i++;
        count++;
        s+=i;
        count++;
    }
    count++;
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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